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Principles of Macroeconomics

Lecture 10: Keynesian Economics and IS-LM Analysis

Prof. Dr. Jan-Egbert Sturm

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Lecture 9: Business cycles

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Business Cycle Models

1. Labour market side – the (new) classical models
2. Labour market side – the (new) Keynesian models
3. Product market side – the (new) classical models
4. Product market side – the (new) Keynesian models
5. Supply side – Real Business Cycle models
   - Production factors determine supply on goods and service markets

- If markets are assumed to clear: (new) classical models (or RBC models)
- If markets are not assumed to clear: (new) Keynesian models

1. Labour market side – the (new) classical models

- Labour markets clear but workers have ***imperfect information***
  - Based around the concept of ***anticipated*** and ***unanticipated*** price changes
- If price rises are ***anticipated*** (i.e. ideal world without this form of business cycle)
  - Workers will recognize that real wages would fall without a nominal wage increase and will supply less labour
  - The demand for labour will be greater than the supply of labour, so nominal wages will rise
  - This will keep the real wage effectively constant at the new equilibrium
- If workers ***do not anticipate*** (and do not realise) the change in prices
  - Real wages fall, firms demand more labour but workers continue to offer the same amount of labour
  - The demand for labour will be in excess of supply and nominal wages will rise but less than prices
  - The result is that output will rise but the real wage will fall (real wages as countercyclical)

2. Labour market side – the (new) Keynesian models

- Keynes believed markets do not clear quickly
  - Markets are impaired by the existence of (sticky prices and) ***sticky wages***
- Nominal wages are often slow to adjust in the economy due to long-term employment contracts
  - An increase in prices will make real wages lower, increasing labour demand and output
    - Real wages are countercyclical

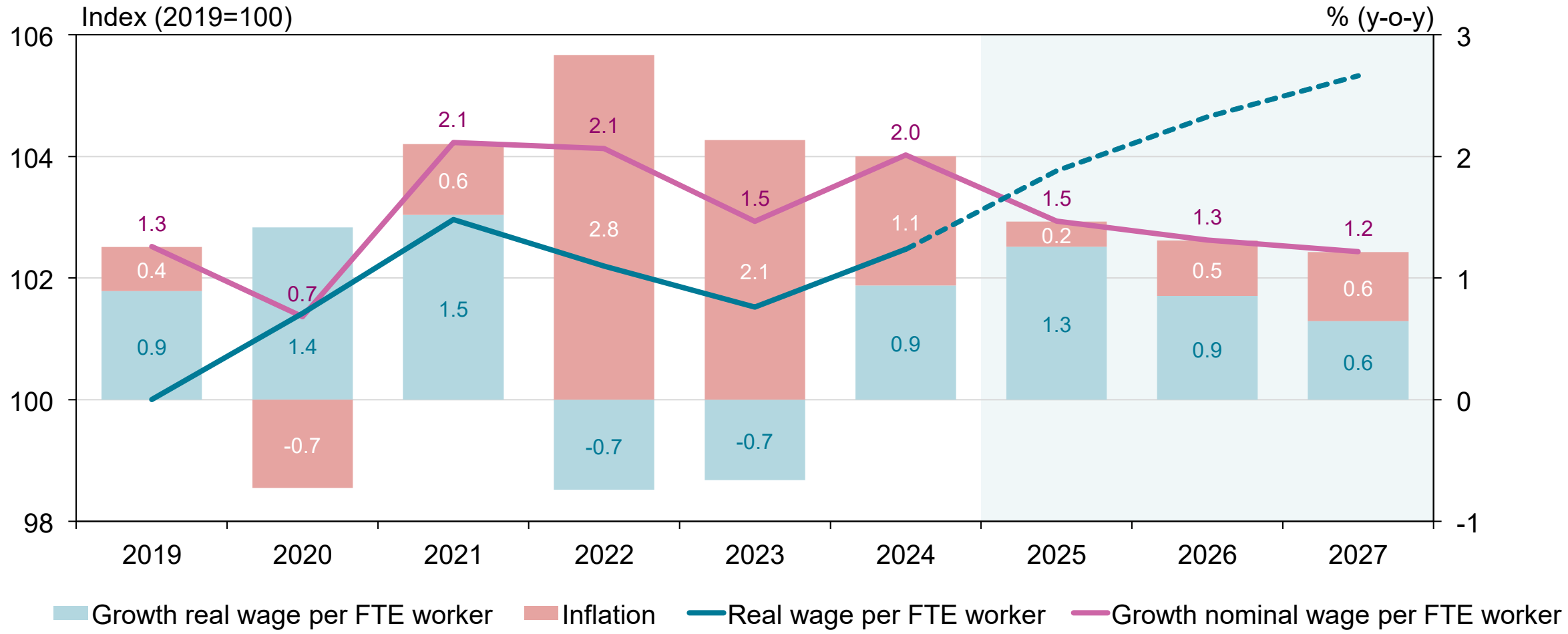
3. Product market side – the (new) classical models

- Information problems might lead to sub-optimal decisions
- Firms might not be able to distinguish between inflation and a relative price change
- An observed increase in the price of the goods they supply might lead them to produce more

New Classical models are driven by asymmetric information

- The New Classical interpretation rests on this ***inflation fallacy***
  - Workers misinterpret a rise in nominal wages as a rise in real wages
  - Firms misinterpret a rise in prices as an increase in demand for their goods
- The economy moves to a temporary equilibrium where the expectations of some economic actors are not fully incorporated because they are incorrect
  - Over time, workers realize that real wages have changed and negotiate for wage rises
  - Firms' costs rise and some firms cut back supply
  - The economy returns to trend output but at a higher price level once expectations have fully adjusted
- Cyclical implications when output is above trend:
  - Unemployment is countercyclical and employment will be above trend and so be procyclical.
  - In case of asymmetric information problems on the labour market, inflation is procyclical and real wages are countercyclical because as output rises real wages fall
  - If there is an information problem on the goods/service market, then real wages move procyclical. The boom creates higher labour demand. Without information problem on the labour market, this implies that this higher demand will lead to the equilibrium real wage to be higher

Wage development in Switzerland – a short-run perspective



Quellen: SECO, KOF (12.10.2025)

4. Product market side – the (new) Keynesian models

- The prices of some goods and services can also be slow to respond to changes in the economy
  - This can be due to menu costs or contractual arrangements
  - Because not all prices adjust instantly to changing conditions,
    an unexpected increase in the price level leaves some firms with lower-than-desired prices
  - This increases sales and cause firms to increase the quantity of goods and services supplied
- In the short run, output increases above trend but prices do not change because of being sticky
- Over time, the economy returns to trend, because firms will eventually be able to raise prices

- Employment, real wages and inflation are procyclical and unemployment is countercyclical

5. Supply side – Real Business Cycle models

- In this model both ***positive and negative technology shocks*** affect productivity
- Assume there are no market imperfections and firms and households are profit and utility maximizing
- A negative technology shock would reduce labour productivity and the demand for labour fall
  - Output also drops and unemployment increases
  - Output falls because aggregate supply falls
  - Prices rise
- When aggregate supply shifts, the expectations of economic actors are still correct
  - (Effectively it is the long-run supply curve that shifts)
- Therefore there is no reason for governments or central banks to intervene
- In real business cycle models, employment, labour productivity and real wages are procyclical

Summary

- Economies experience periods of changing levels of economic activity
- Key macroeconomic data is often time-series data which raises questions about the validity and reliability of statistical tests applied to data sets
- Economic growth appears to follow a trend which rises over a period of time
- Deviations from this trend are known as the business cycle
- The business cycle has characteristic features such as peaks in economic activity, slowdown, troughs and upturns with key turning points
- Trends can be stationary and nonstationary. Nonstationary, data can exhibit deterministic trends which change by a constant amount independent of time or be stochastic where the trend variable changes by a random amount
- Macroeconomic data can be viewed in pairs with one of the pairs being GDP
  – The variable compared may be procyclical or countercyclical
- Collecting macroeconomic data allows economists to view indicators as leading, coincident or lagging

Summary

- Changes in the business cycle can be caused by changes in household and firms' decision making, by external shocks, government policy and changes in confidence or expectations of the future
- Business cycle models differ in the assumptions they make about the extent to which markets clear and the relationship between the supply side and demand side of the economy
- Real business cycles emphasize the effects of changes in technology as causes of changes in economic activity

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Principles of Macroeconomics

Lecture 10: Keynesian Economics and IS-LM Analysis

Prof. Dr. Jan-Egbert Sturm

John Maynard Keynes (1935-36): The General Theory of Employment, Interest and Money

- Keynes' primary message was that recessions and depressions can occur because of inadequate aggregate demand
- He was a critic of classical economic theory, because it could explain only the long-run effects of policies.
  - "In the long run we are all dead.
    Economists set themselves too easy, too useless a task if in tempestuous seasons they can only tell us that when the storm is long past, the ocean will be flat."

- Keynesian economics assumes prices are sticky
  - In what follows, we will (initially) assume prices to be constant

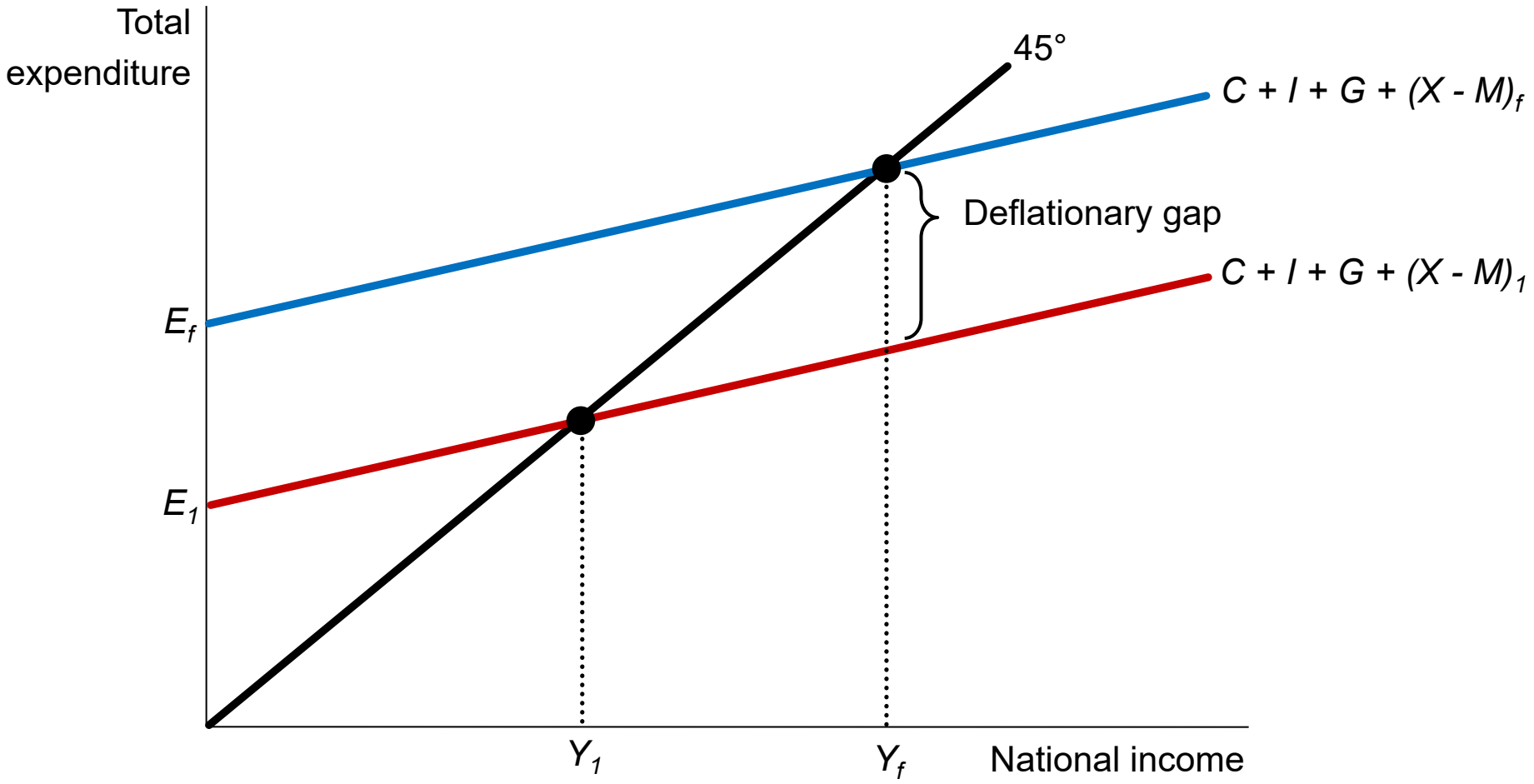
Planned and actual spending

- Keynesian analysis makes a crucial distinction between planned and actual behaviour
  - You may plan to save €100 per month, but if your overtime is cut, then you may actually save only €20 per month
  - A manufacturer may plan to maintain a stock of 50,000 of its widgets, but if sales slow, it may actually find it has 150,000 widgets in stock.
- ***Planned*** spending, saving or investment is the intended actions of households and firms
- ***Actual*** spending, saving or investment is the realized outcome resulting from actions of households and firms
- ***Full employment*** is when those people who want to work at the going rate are able to find a job
- If planned and actual outcomes can be very different, then it is possible for the equilibrium level of national income to be below the full employment level

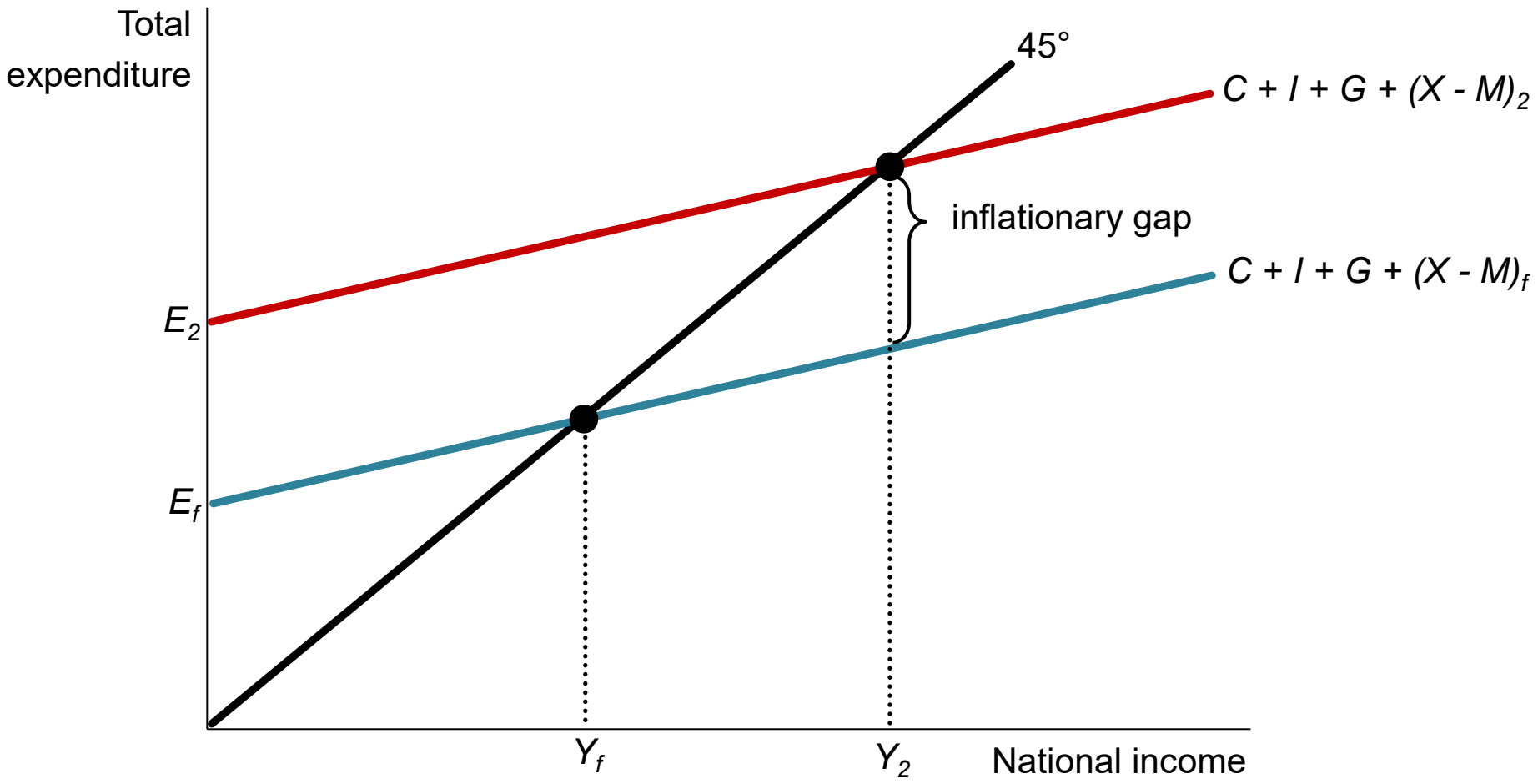
Equilibrium of the Economy

- In the Keynesian cross diagram, the 45° line connects all points where total spending would be equal to national income
  - It is equivalent to the capacity of the economy
- Short-run equilibrium occurs where the expenditure function
  - $E = C + I + G + NX$
  - crosses the 45° line and refers to a point where actual spending is equal to planned spending
- Autonomous expenditure:
  - spending that is not dependent on income/output
- To eradicate an inflationary or deflationary gap a government can influence the components of aggregate demand through fiscal and monetary policy bringing about an equilibrium that is closer to the full employment level of national income
  - The ***deflationary*** or ***output gap*** is the difference between full employment output and expenditure when expenditure is less than full employment output
  - The ***inflationary gap*** is the difference between full employment output and actual expenditure when actual expenditure is greater than full employment output

Keynesian cross: Deflationary gap



Keynesian cross: Inflationary gap



The Multiplier Effect and the Accelerator Principle

- The ***multiplier effect*** refers to the additional shifts in aggregate demand that result when expansionary fiscal policy increases income and thereby increases consumer spending
- The expenditure function E = C + I + G + NX
  - The slope of the expenditure function is dependent on the multiplier effect
- Government purchases are said to have a ***multiplier effect*** on aggregate demand
  - A euro of government purchases can generate more than a euro of aggregate demand
- The multiplier effect can be strengthened by response of investment to higher levels of demand
  - The ***accelerator principle*** relates the rate of change of demand to the rate of change in investment
    - Suppose a machine can produce 1,000 DVDs per week. Current demand is 800 DVDs per week
    - A 10% rise in demand can be accommodated with no new investment in machinery
    - Suppose the following year demand rises by 20%, taking demand to 1,056 units
    - The firm may decide against investing in new capacity and raise prices in face of the shortage
    - But if demand rises by 25% the firm is very likely to invest in a new machine
    - An increase in demand of 25% has led to an ‘accelerated’ rise in investment of 100%

A Formula for the Spending Multiplier

- The ***marginal propensity to consume*** (MPC)
  - It is the fraction of extra income that a household consumes rather than saves
  - Multiplier = 1/(1 - MPC) = 1/MPS
    (where MPS is the marginal propensity to save)
- If the MPC is 3/4, then the multiplier will be:
  - Multiplier = 1/(1 - 3/4) = 4
    - Government pays €10 billion to Nucelec to build three new nuclear power stations
    - Nucelec employees and contractors receive increased income and spend it on consumer goods
    - In this case, a €10 billion increase in government spending on power plants generates €40 billion of increased demand for goods and services

Other Applications of the Multiplier Effect

- The logic of the multiplier effect is not restricted to changes in government purchases
- It applies to any event that alters spending on any components of GDP – consumers' expenditure, investment, net exports, or government spending
- ***Autonomous expenditure***: spending that is not dependent on income/output
  - Any increase in autonomous expenditure will lead to a multiplier effect
  - The additional spending that follows as a result of the multiplier effect is termed ***induced expenditure***
- The multiplier is important because it shows how the economy can amplify the impact of changes in spending
  - A small initial change in C, I, G or NX can end up having a large effect on national income
  - The eventual change in income is determined by the marginal propensity to consume (MPC)

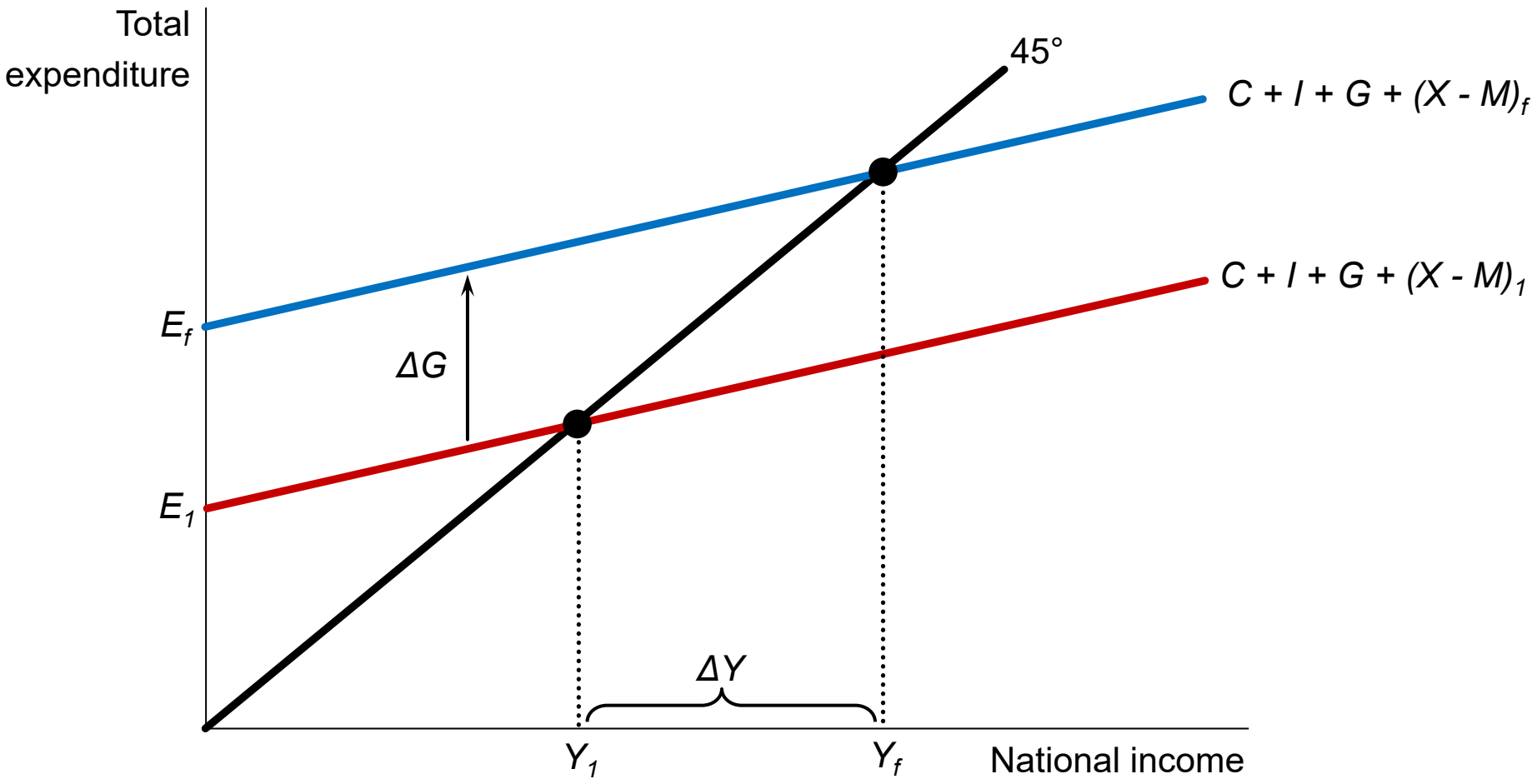
Other Applications of the Multiplier Effect

- In an open economy any extra income is not simply either spent or saved – some of the extra may be spent on imports or go to the government in taxes
  - Savings, imports, and taxes are classified as withdrawals (W) from the circular flow of income and are ***endogenous*** as they are directly related to changes in income
    - MPT: the marginal propensity to taxation
    - MPM: the marginal propensity to import
    - MPS: the marginal propensity to save
- The multiplier would now be expressed as: **Multiplier = 1/(MPS + MPT + MPM) = 1/MPW**
- The MPW will reduce the size of the multiplier and thus the impact on national income of any increase in autonomous expenditure
- The higher the MPW the smaller the multiplier and the shallower/flatter the slope of the expenditure line

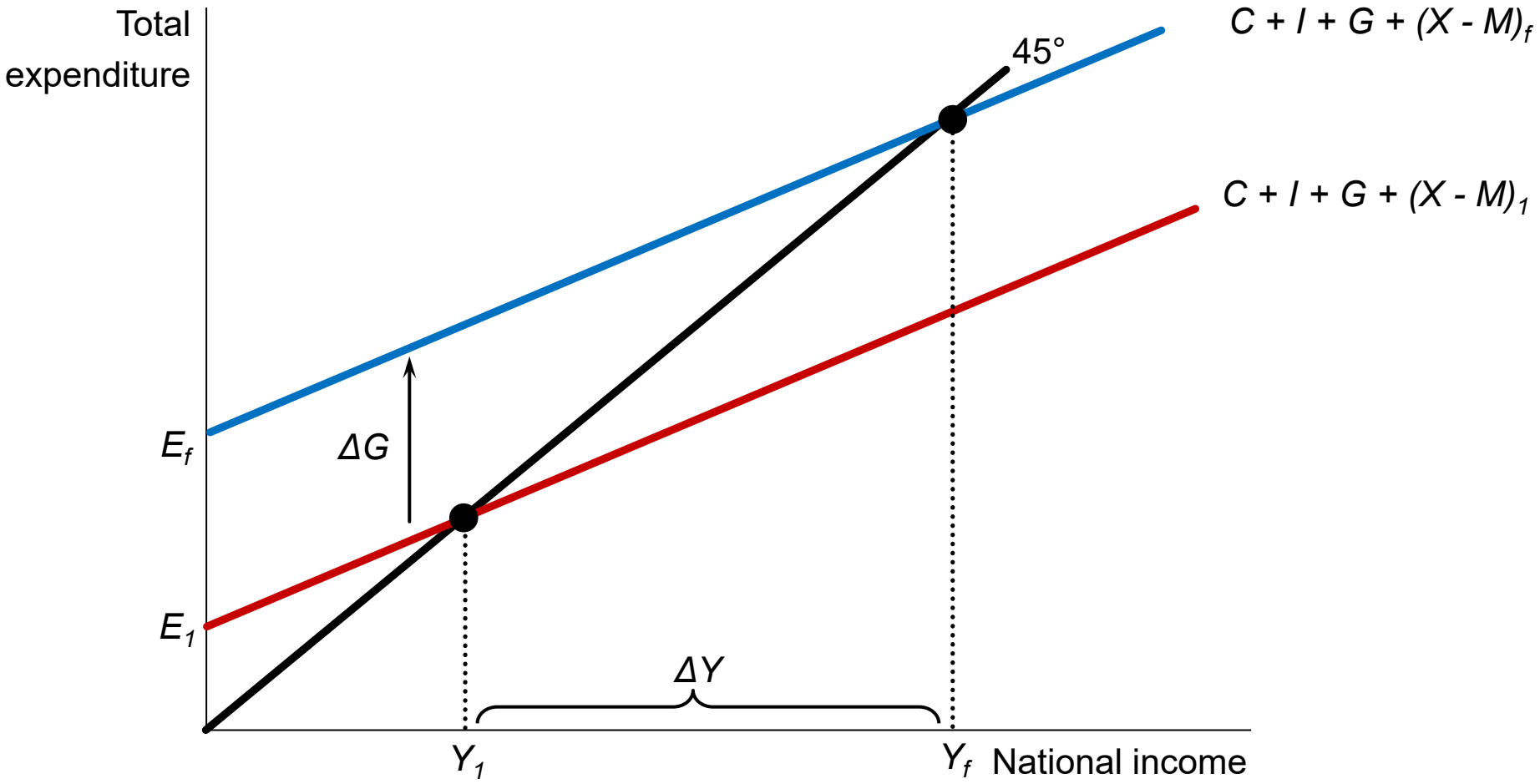
Mathematical example

- Assume that consumption, taxation and imports depend linearly on income
  - $C = C_0 + (1 - mps) * Y - T$
  - $T = T_0 + mpt * Y$
  - $M = M_0 + mpm * Y$
- Assume investment, government spending and exports exogenously given
- This implies (given $E = C + I + G + X - M$)
  - $E = C_0 + (1 - mps) * Y - (T_0 + mpt * Y) + I_0 + G_0 + X_0 - (M_0 + mpm * Y\ )$
  - $E = C_0 + (1 - mps) * Y - T_0 - mpt * Y + I_0 + G_0 + X_0 - M_0 - mpm * Y$
  - $E = (C_0 - T_0 + I_0 + G_0 + X_0 - M_0) + \big((1 - mps) - mpt - mpm\big) * Y$
  - $E = E_0 + (1 - mps - mpt - mpm) * Y = E_0 + (1 - mpw) * Y$
  - In equilibrium $Y = E \rightarrow Y = {}^{1}/_{mpw} * E_0$

Keynesian cross: Deflationary gap



Keynesian cross: Deflationary gap – higher multiplier



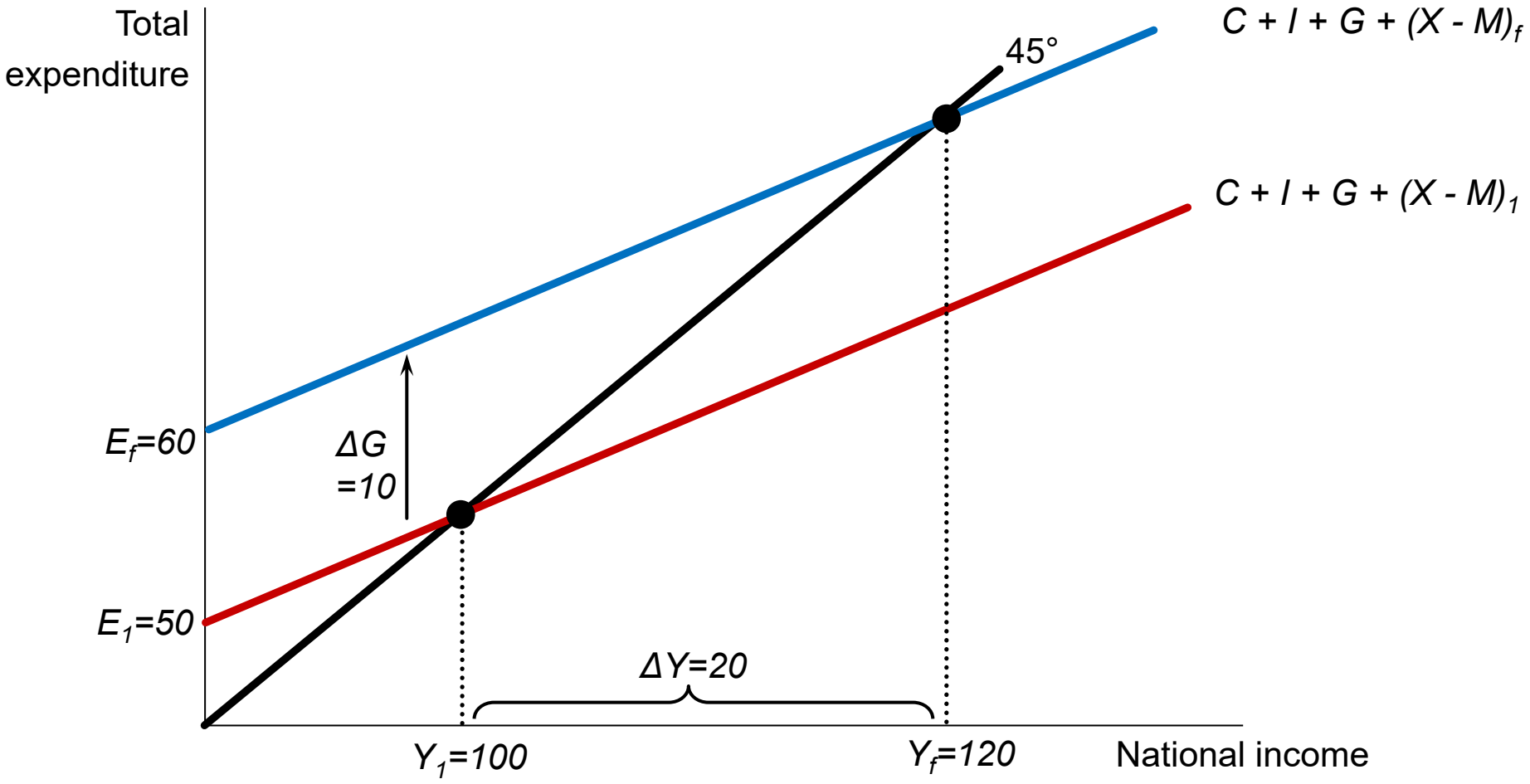
Equilibrium of Planned Withdrawals and Injections

- In equilibrium planned expenditure equals actual income
  - This equilibrium is referred to as equilibrium in the goods market as all output is bought by households and firms
  - However, if actual withdrawals were greater than planned injections then the economy would be experiencing a deficiency in demand
  - Then a government may decide to increase its spending and income will rise by even more because of the multiplier

Example: Equilibrium of Planned Withdrawals and Injections

- Assume full employment output (Y) is 120 billion
- Assume private saving, $S = 0.1Y$; taxes, $T = 0.2Y$; imports, $M = 0.2Y$
- Assume investment (I) was 20 billion, government spending (G) 20 billion and exports (X) 10 billion (Hence, total injection is $I + G + X = 50$ billion)
- Note: $Y = C + I + G + X - M$
  $\leftrightarrow (Y - C - T) + T + M = I + G + X$
- Planned $S + T + M =$ Planned $I + G + X$
  or $0.1Y + 0.2Y + 0.2Y = 0.5Y = 50$ billion, so $Y = 100$ billion ($multiplier = 2$)
  - The deflationary gap is 20 billion
- The government decides to spend an extra 10 billion: $(I + G + X) = 60$ billion
- Now $0.5Y = 60$ billion so Y rises through the multiplier to 120 billion

Keynesian cross: Deflationary gap – higher multiplier

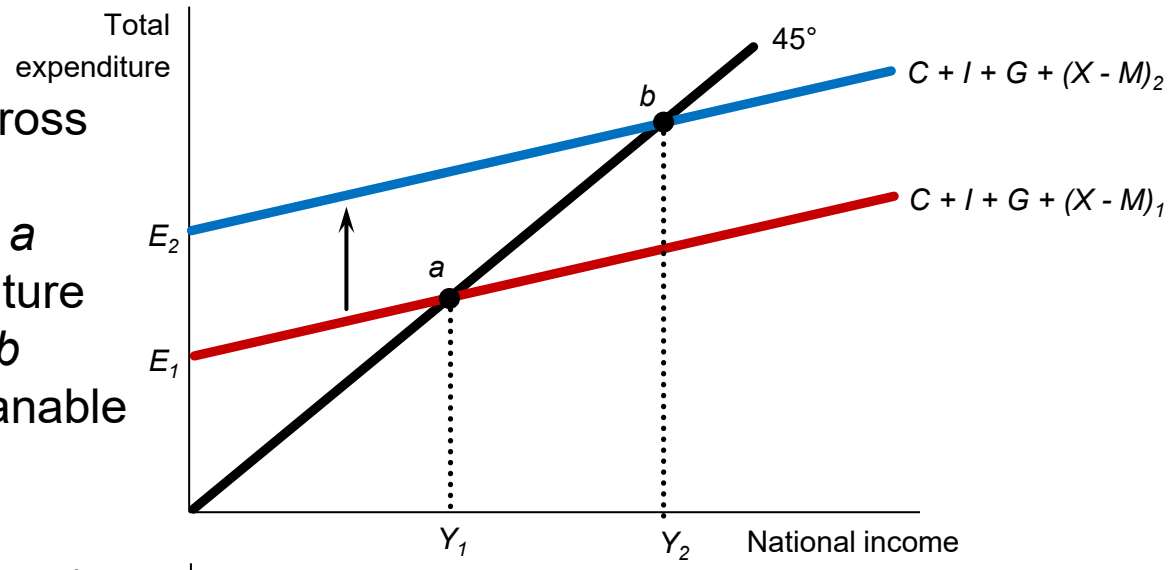


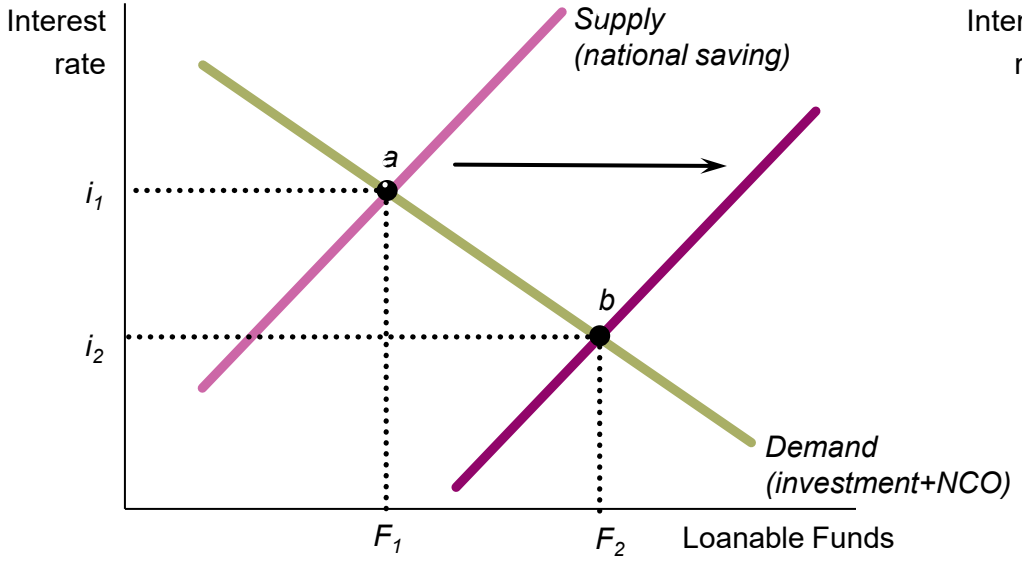
The IS and LM Curves

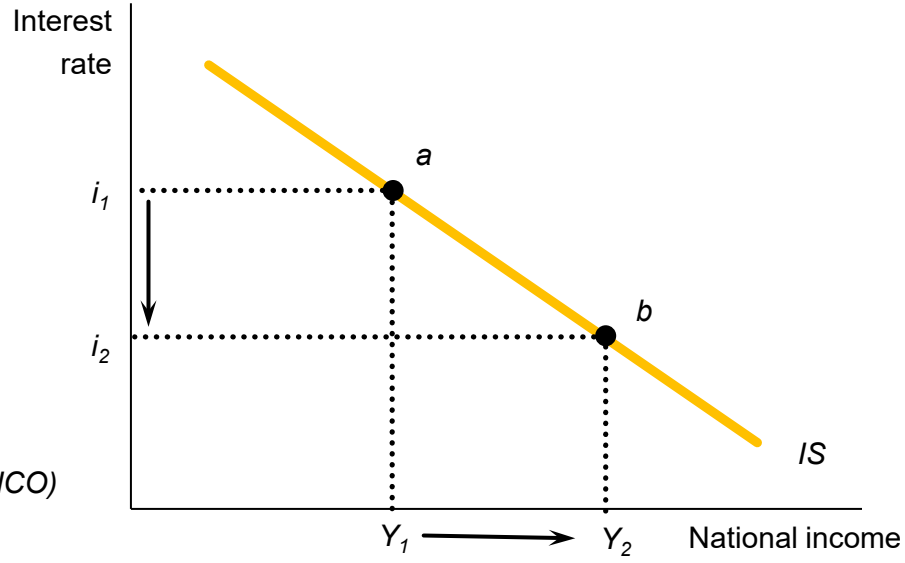
- IS-LM describes equilibrium in goods market and the money market and together determines general equilibrium in the economy
  - IS – Investment and Saving
  - LM – Liquidity and Money
- What links the goods and money markets is the rate of interest i

The IS Curve

- The IS curve is derived from the Keynesian cross diagram
  - In the upper panel, the initial equilibrium is *a*
  - A fall in the interest rate raises the expenditure line and a new equilibrium occurs at point *b*
- The IS curve can also be derived from the loanable funds market perspective





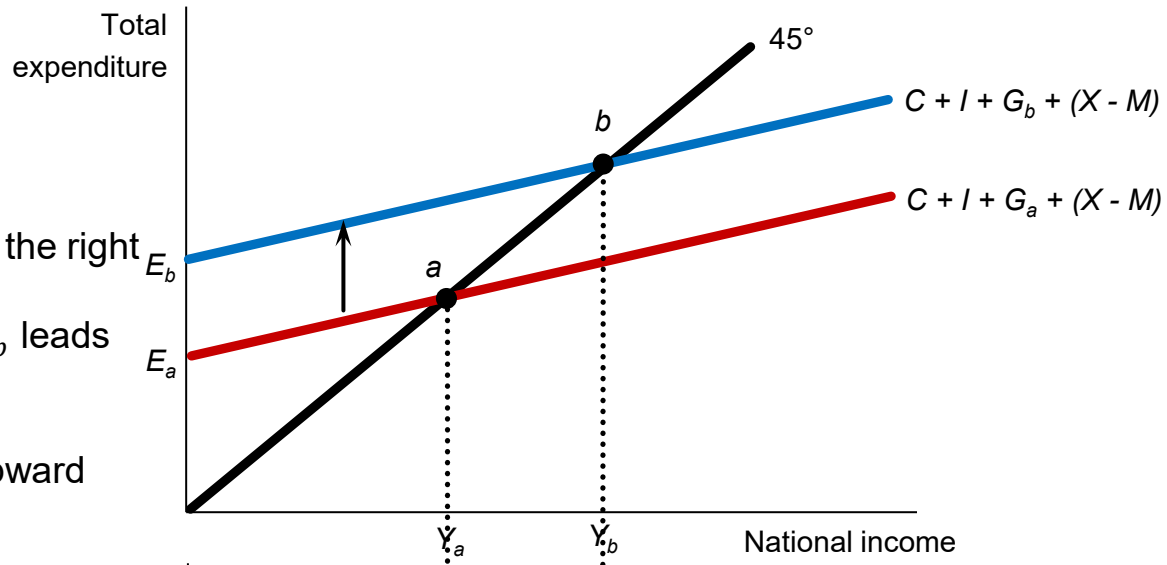


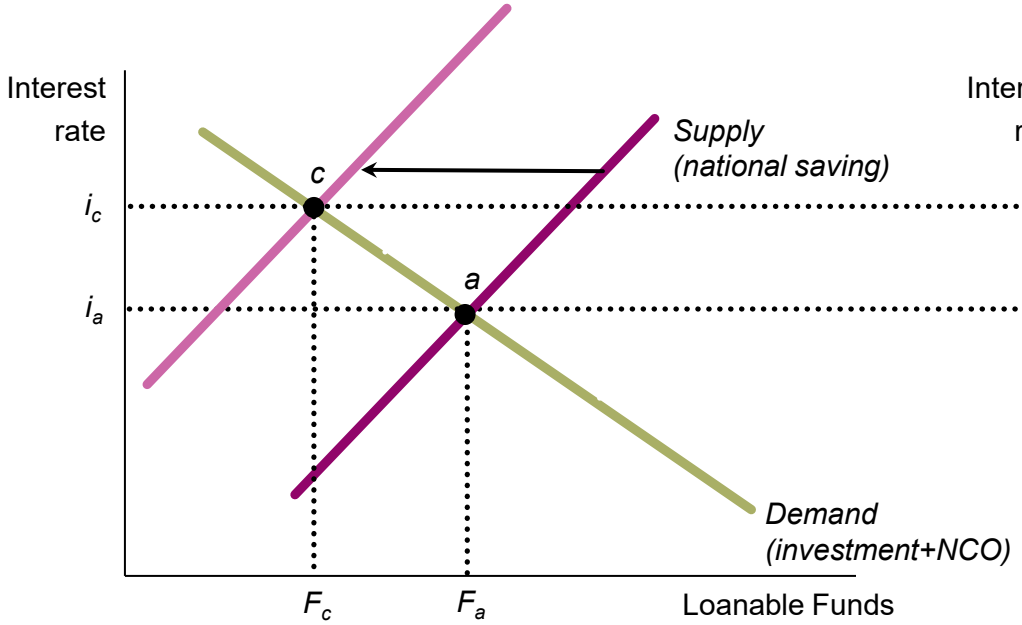
The IS Curve

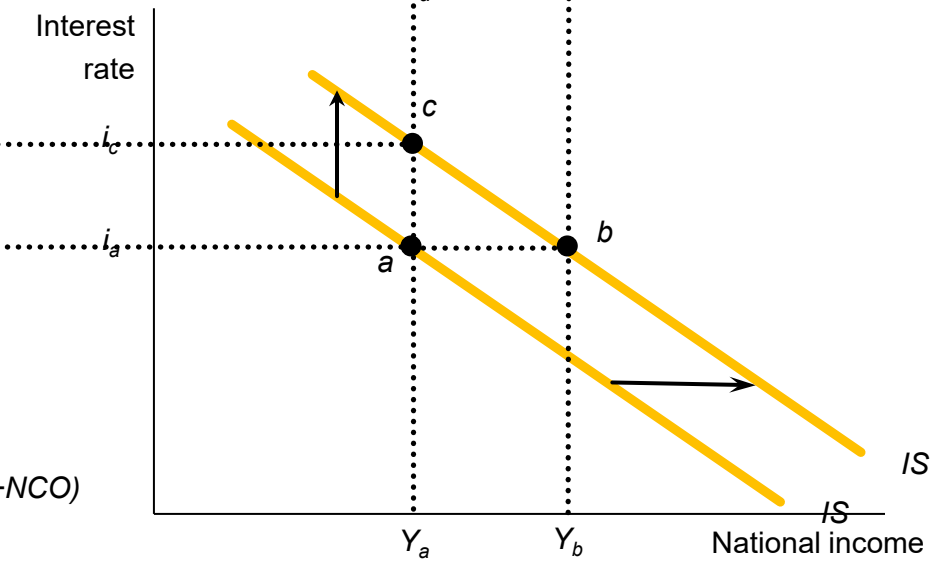
- The IS curve shows an inverse relationship between the interest rate and output.
- Note that the slope of the IS curve depends on how responsive consumption and investment expenditures are to changes in interest rates and on the size of the multiplier.
- The more responsive are *C* and *I*, the flatter is the IS curve.
- Shifts in the IS curve come about as a result of changes in autonomous expenditure.
- A rise in government spending, independent of any change in interest rates, would lead to a shift of the IS curve to the right.
- A fall in exports would lead to a shift of the IS curve to the left.

Changing government spending and its consequences for the IS Curve

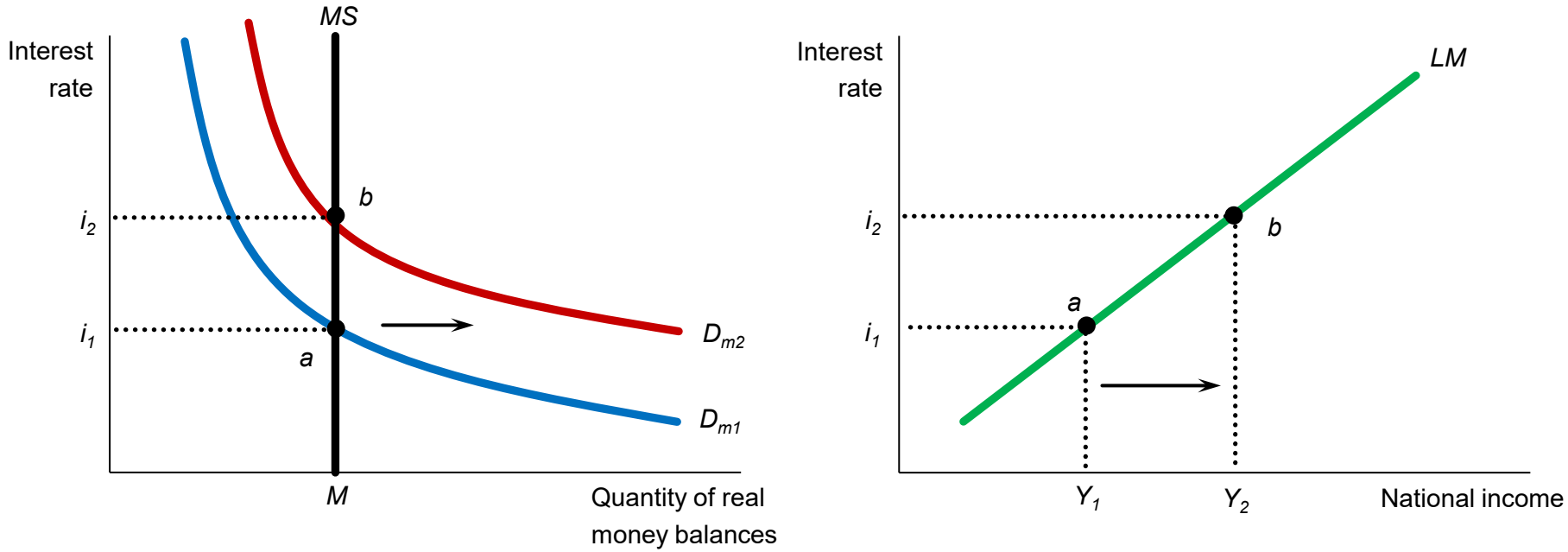
- Keynesian-cross perspective:
  - For a given interest rate (i_a), an increase of G_a to G_b
    - leads to an increase of Y_a to Y_b
  - In the IS diagram this implies that the IS curve shifts to the right
- Loanable-market perspective:
  - For a given level of income (Y_a), an increase of G_a to G_b leads to lower national savings
    - pushing the interest from i_a to i_c
  - In the IS diagram this implies that the IS curve shifts upward





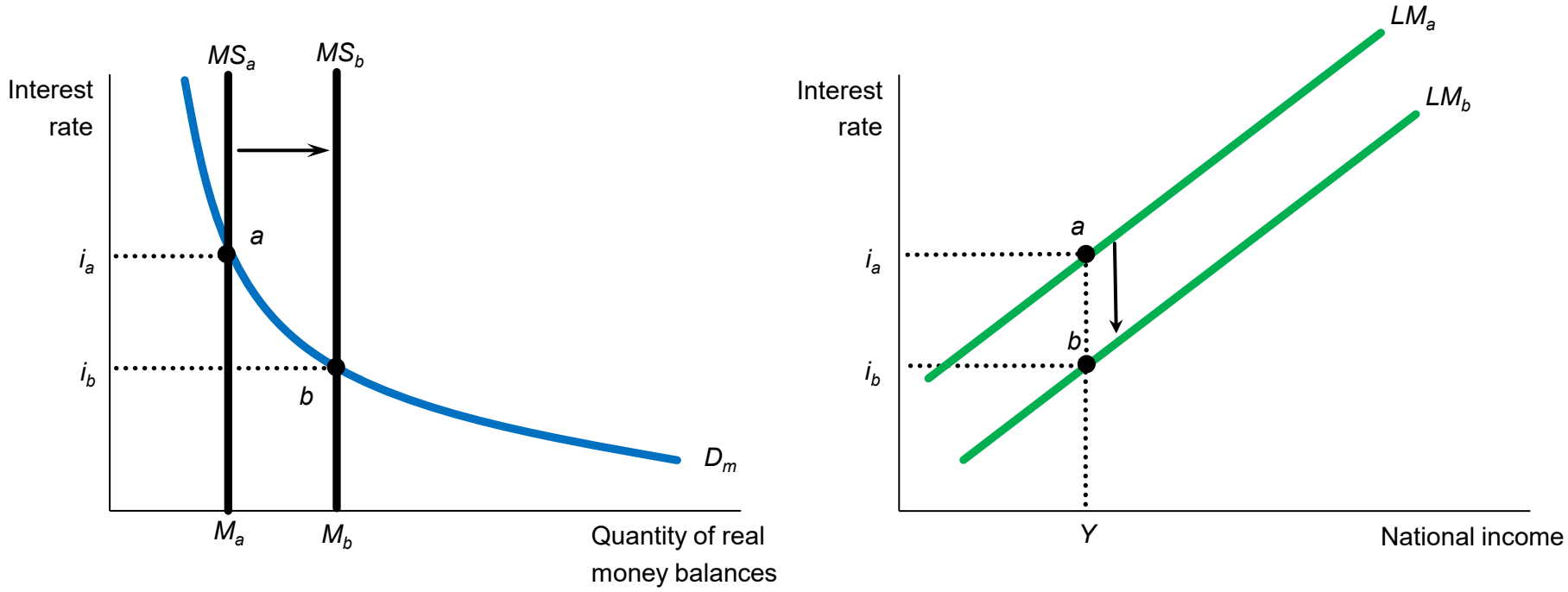


The LM Curve



- The LM curve is derived from our money market diagram
  - In panel (a) initial equilibrium is point *a*
  - A rise in national income increases the demand for money and a new equilibrium occurs at point *b*
- Positive slope indicates that an increase in income is associated with an increase in the interest rate
  - The slope depends on how responsive the demand for money is to changes in interest rates
  - The LM curve shifts to the right/left if the central bank expands/contracts money supply

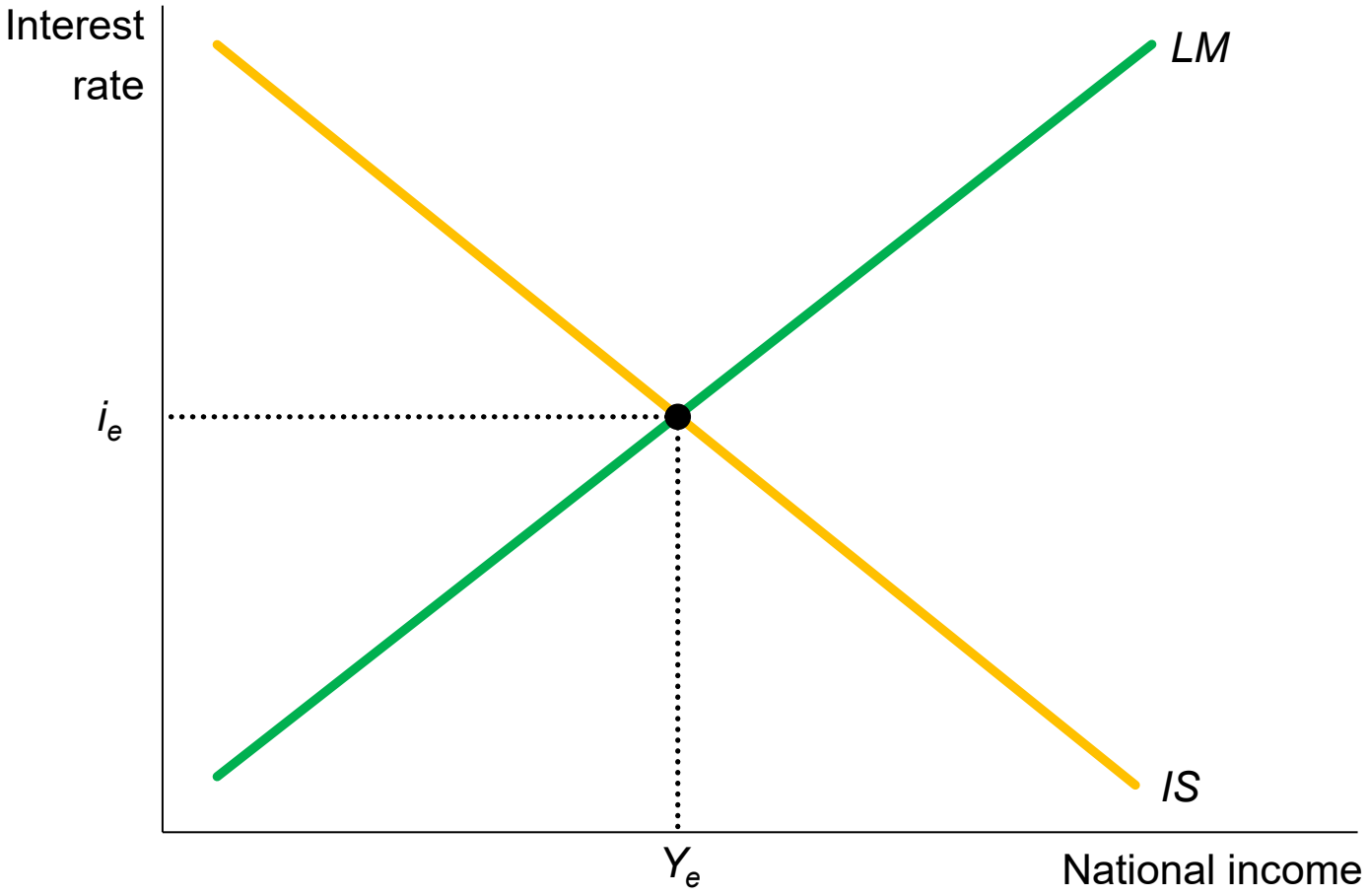
Changing money supply and its consequences for the LM Curve



- If the central bank increases money supply from Ma to Mb, then (for a given money demand) the interest rate has to fall. Only for a lower interest rate society is willing to hold the additional money (assuming that e.g. income is unchanged)
- In the LM diagram this implies that the LM curve shift downward (or to the right).
  - For the same level of income (Y), the interest rate will now be lower.
  - (or: to keep the interest rate constant, income needs to increase (moving the money demand curve outward/upward))

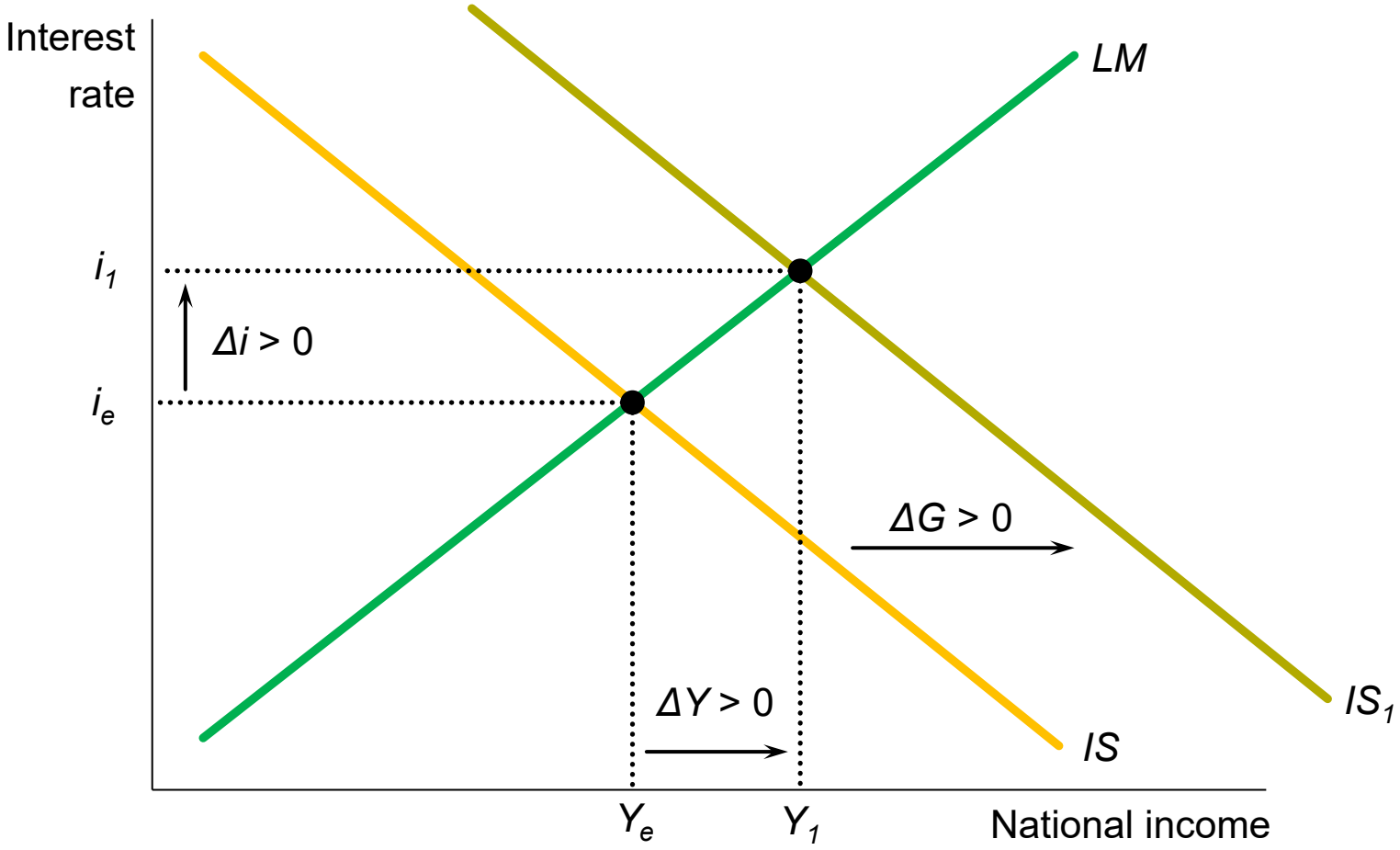
General Equilibrium Using the IS-LM Model

- Equilibrium in the IS-LM model is found where the IS curve intersects the LM curve
- Recall that each point on the:
  - IS curve represents a point of equilibrium in the goods market
  - LM represents a point of equilibrium in the money market
- So a point that lies on both curves represents equilibrium in both the goods market and the money market



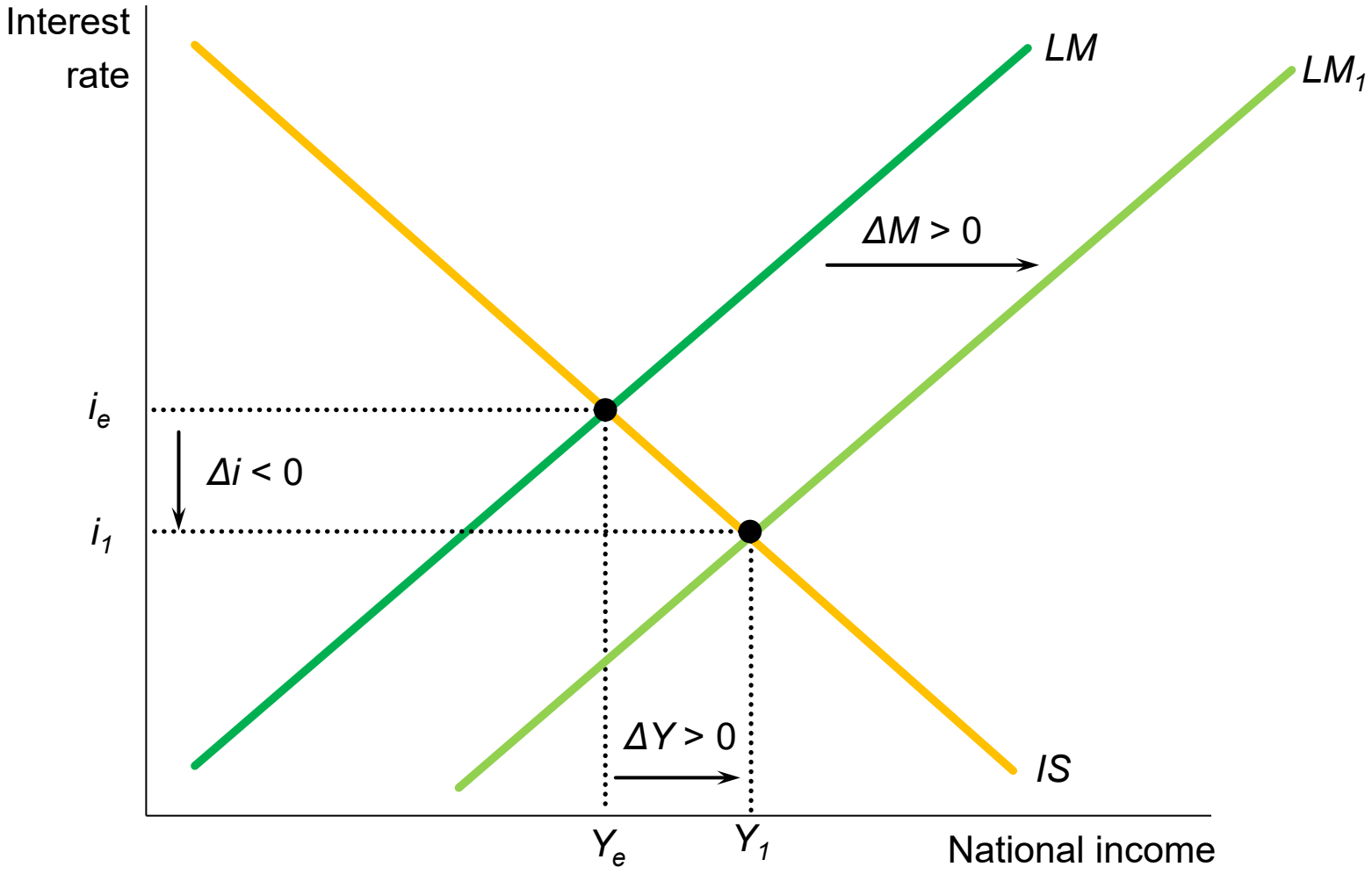
The Effect of a Change in Fiscal Policy

- Suppose the government chooses to increase its spending to boost economic activity
  - This increase in autonomous expenditure shifts the IS to the right
  - There will be an increase in national income but also an increase in interest rates



The Effect of a Change in Monetary Policy

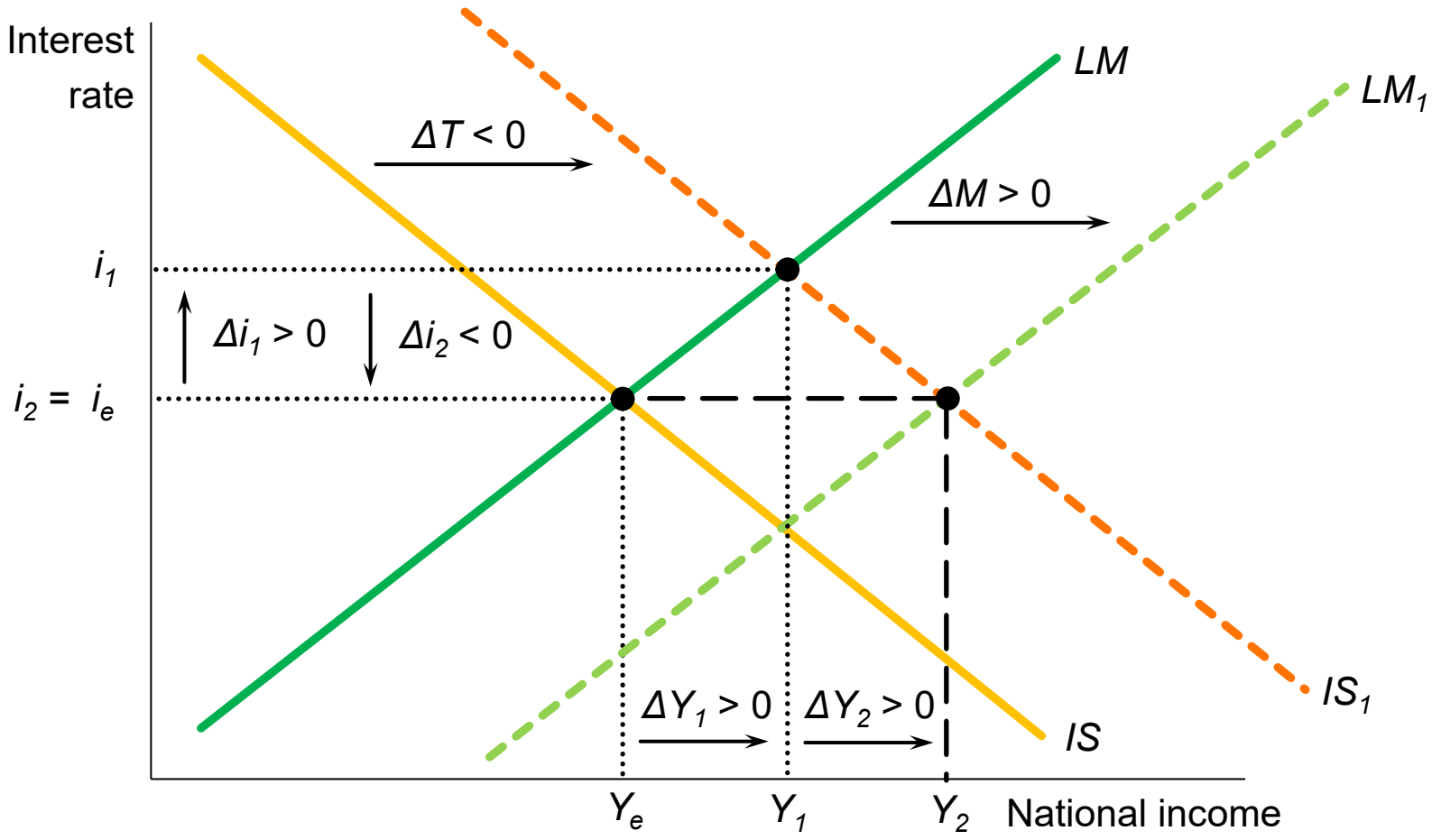
- The central bank decides to expand the money supply
  - The LM curve will shift to the right
  - There will be an increase in national income and also a fall in interest rates



The Effect of a Change in Monetary Policy

- In reality, central banks do not act totally in isolation of government even if they are independent
  - Central banks monitor fiscal policy to see what effects they might have on inflationary pressures in the economy
- Suppose the government reduces taxation to boost economic activity
  - The IS curve will shift to the right and both national income and interest rates will rise
- If the central bank wants to keep interest rates constant then it must expand the money supply and so shift the LM curve to the right

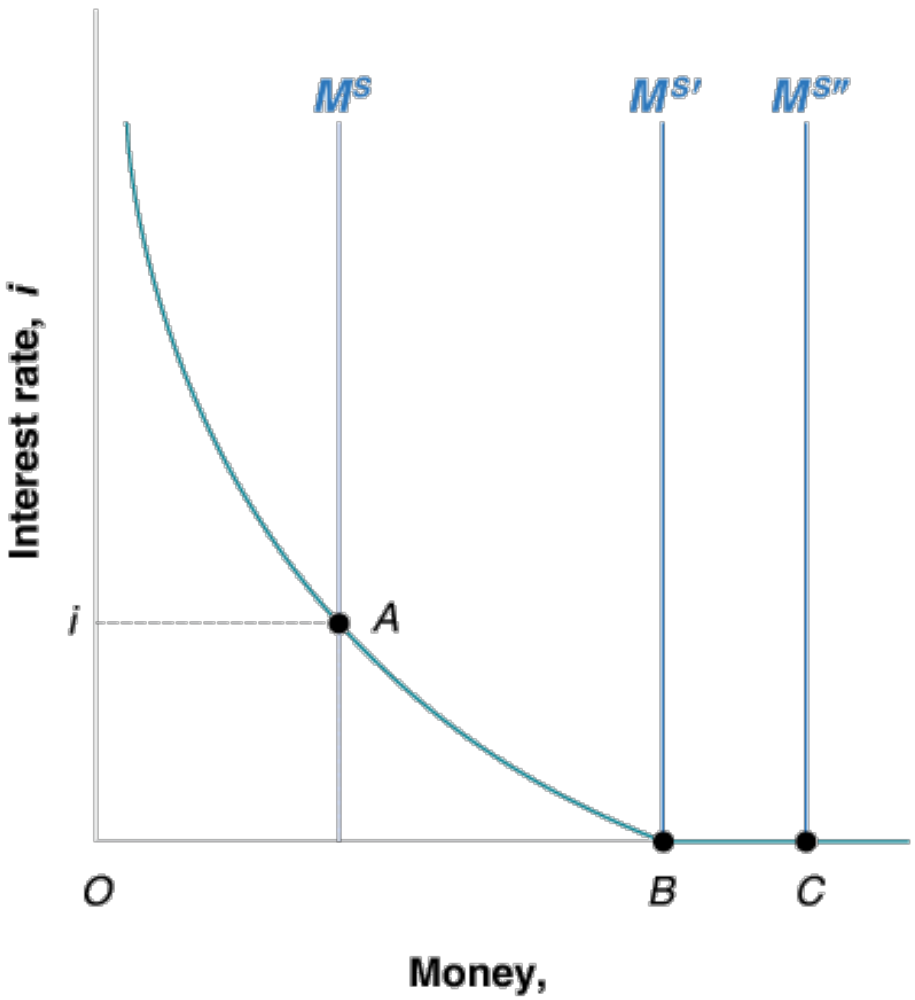
General Equilibrium



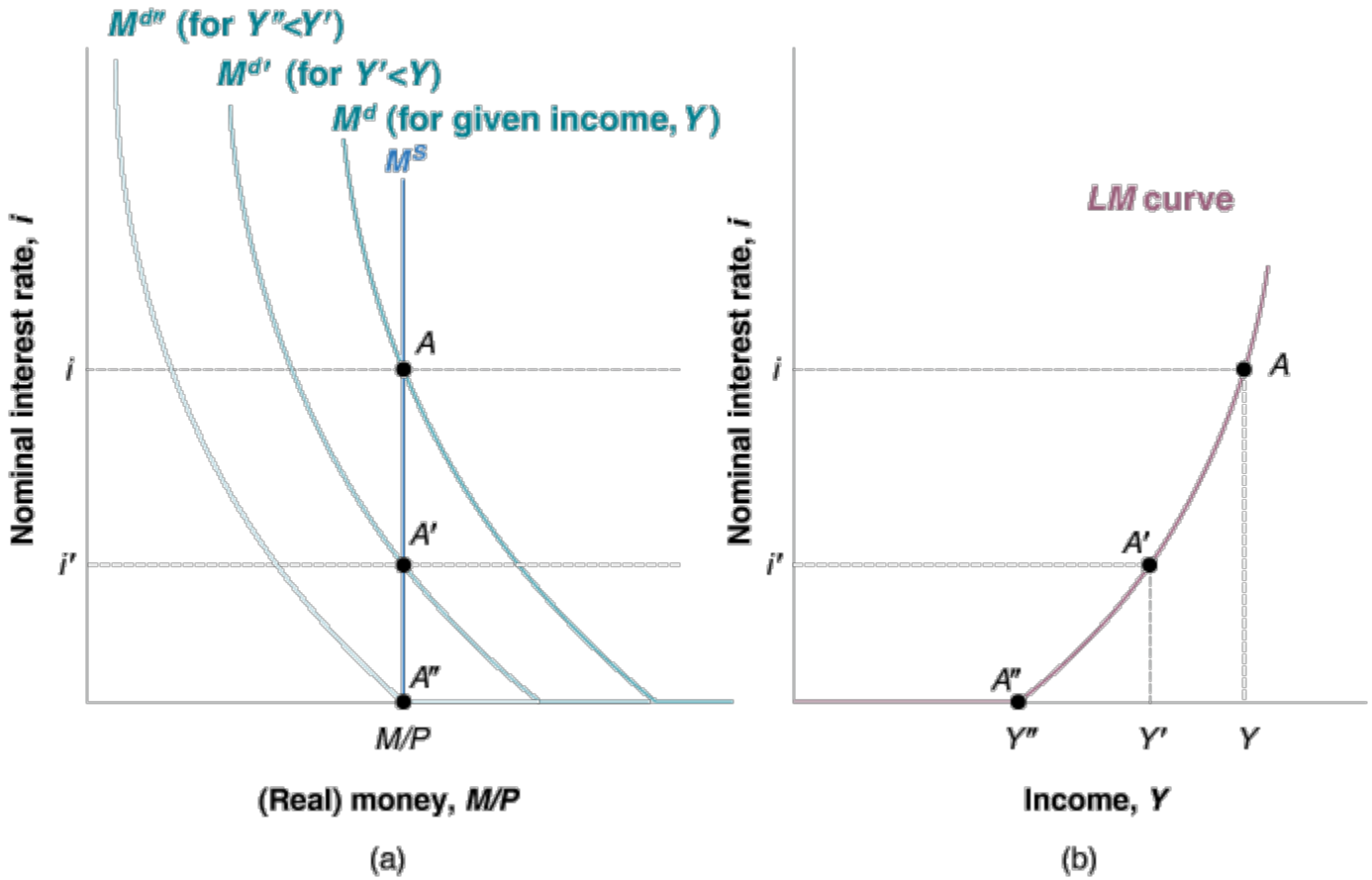
The Liquidity Trap

- When the nominal interest rate is zero, and people have enough money for transaction purposes, they become indifferent between holding money and holding bonds as the return is the same (i.e. 0)
  - The demand for money becomes horizontal
  - Increases in money supply do not effect the nominal interest rate

- Consider the effects of an increase in money supply:
  - Starting from an equilibrium at point A
    - An increase in the money supply leads to a lower interest rate
  - Now consider the case where the money supply is at B
    - An increase in money supply has no effect on the interest rate
    - As the interest rate becomes equal to zero, people want to hold an amount of money ***at least*** equal to the distance OB, which is what they need for transaction purposes

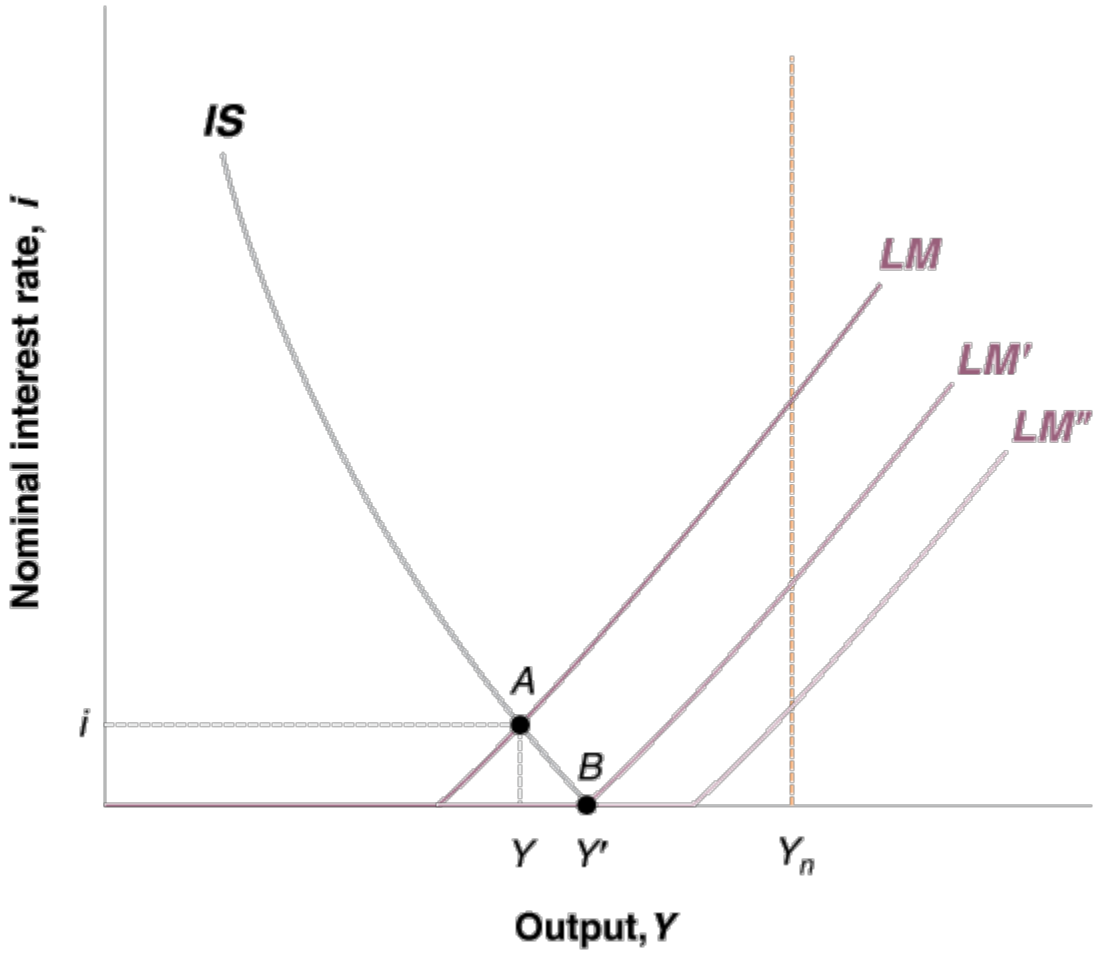


Deriving the LM Curve (allowing for a liquidity trap)



The Liquidity Trap

- In the presence of a liquidity trap, there is a limit to how much monetary policy can increase output
- Monetary policy may not be able to increase output back to its natural level

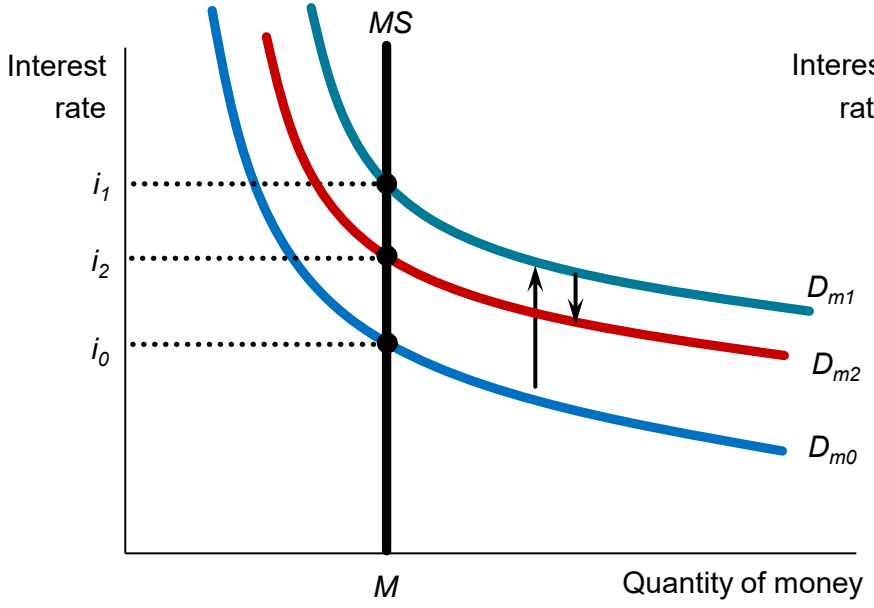


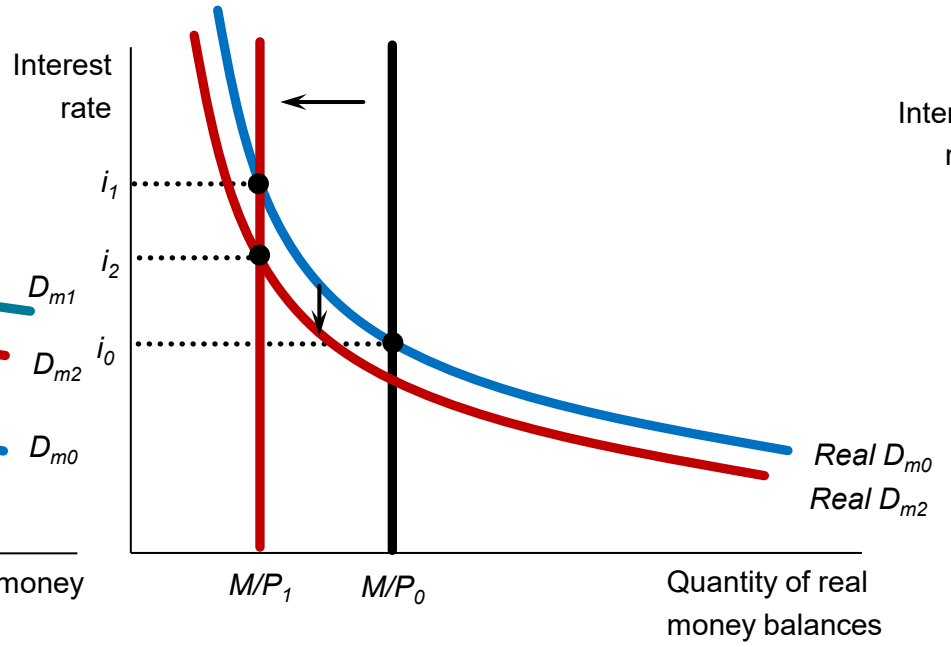
From IS-LM to Aggregate Demand

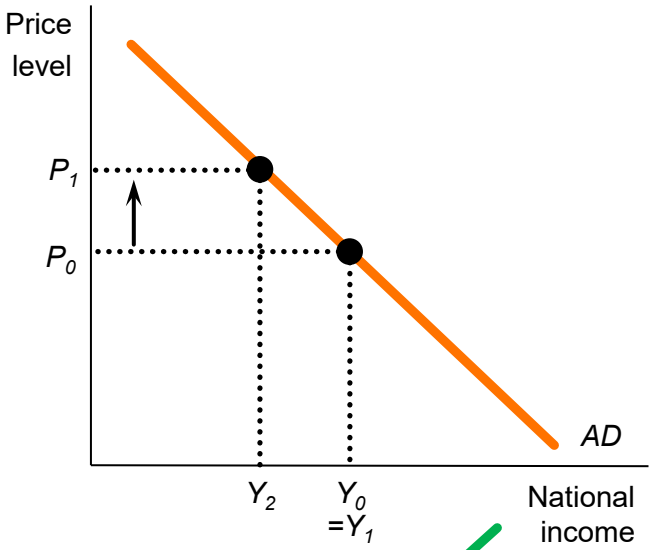
- It is a small step from the IS-LM model to the aggregate demand and aggregate supply model
  - Aggregate demand shows the relationship between national income and the price level
- Note that we have so far assumed prices to be stable/constant/sticky
  - Hence, P was fixed
  - We have therefore been referring to the real money supply – what money can actually buy
  - Real money supply (M/P) is given by M (nominal money supply) divided by P, the price level
- A rise in P causes a fall in M/P, shifting the LM curve to the left
- The result is that national income falls and interest rates rise
- Hence, higher P lowers aggregate demand

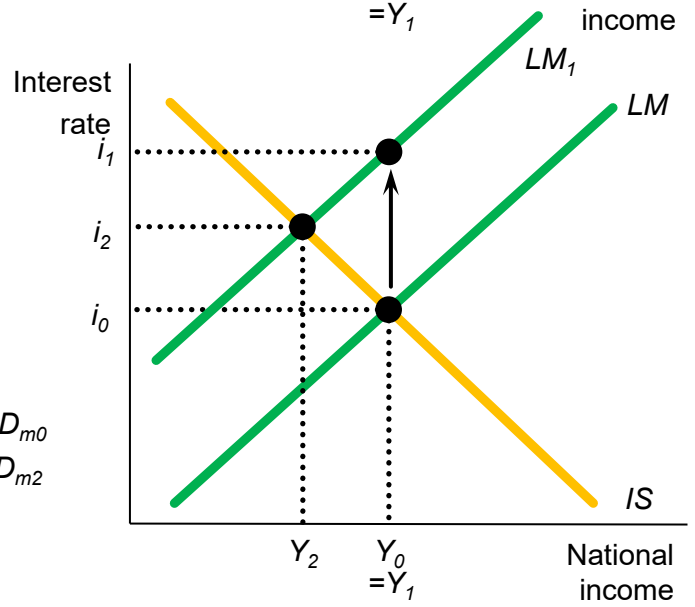
General Equilibrium

1. A rise in the price level reduces real money balances and shifts the LM curve to the left
   - This raises the equilibrium interest rate and lowers national income
2. The aggregate demand curve AD shows the inverse relationship between the price level and national income





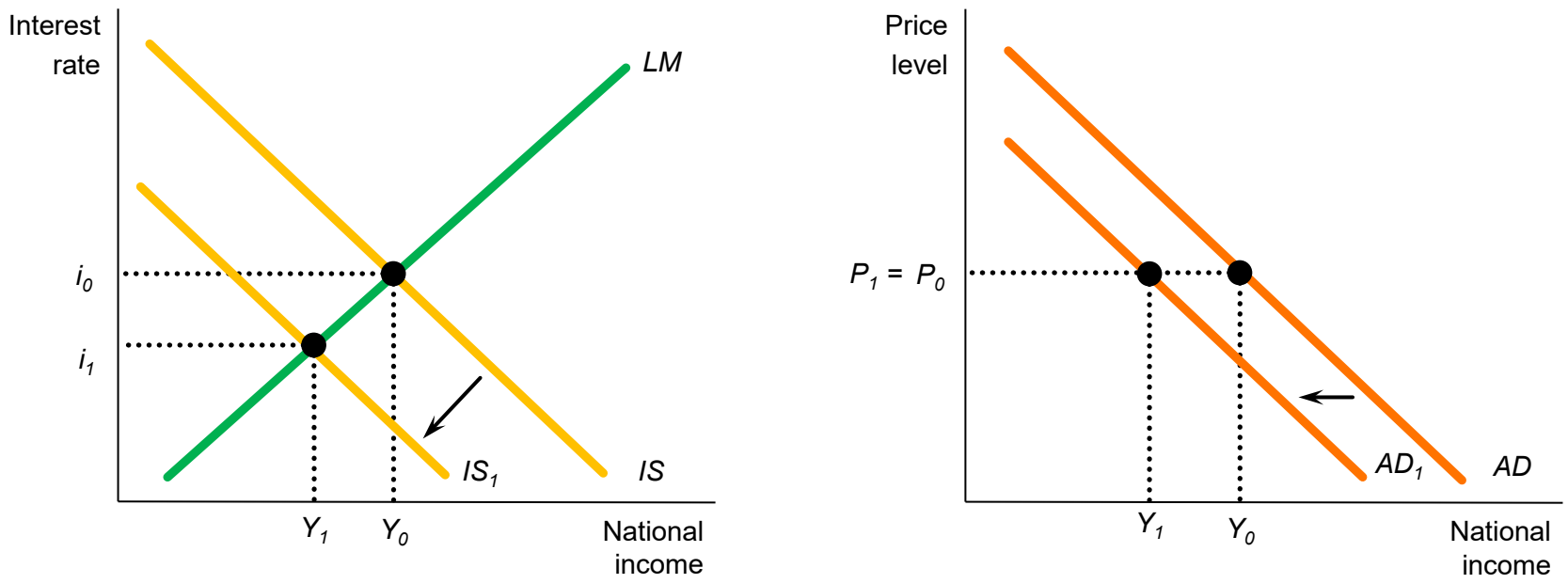




From IS-LM to Aggregate Demand

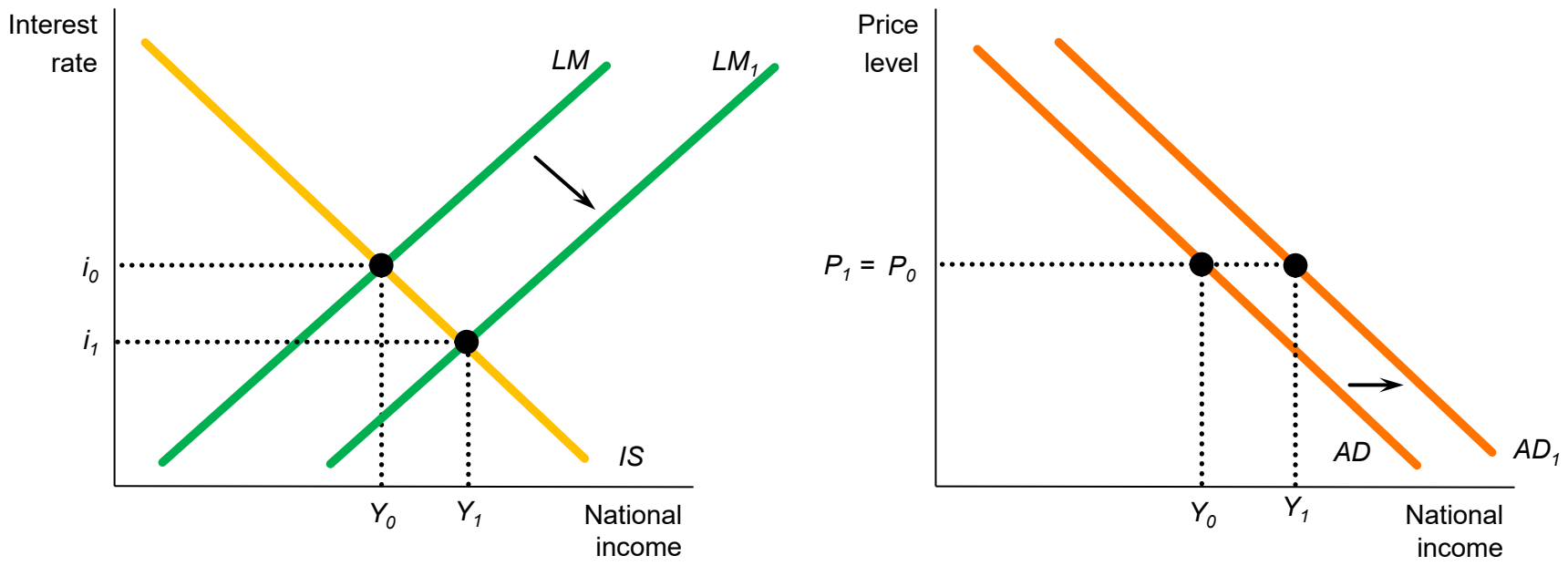
- Assume the price level remains constant, then a change in national income in the IS-LM model will result in a shift in the aggregate demand curve
- For example:
  - Austerity measure of tightening fiscal policy causes a fall in national income as aggregate demand shifts inwards
  - Central bank loosens money supply, e.g. through quantitative easing, boosts national income as aggregate demand shifts outwards

A Government Tightening Fiscal Policy



- In the short term, tightened fiscal policy reduces national income and the economy's interest rate, but the price level (is assumed to) remain(s) constant

Central Bank Loosens Monetary Policy



- In the short term, expansive monetary policy increases national income and reduces the economy's interest rate, but the price level (is assumed to) remain(s) constant

A Return to Keynesianism?

- The severity of the global recession, the euro crisis and the Corona pandemic has led to calls for fiscal stimulation especially as monetary policy has failed to stimulate growth
- Two views
  - To raise the money governments will either have to tax its citizens more or increase borrowing and therefore crowding out private investment
  - The depth of some of these economic crises was/is such that a fiscal stimulus was/is necessary to get out of a liquidity trap and, far from crowding out, fiscal stimuli would lead to ‘crowding in’